

DAA Output Of Five Experiments

1. Implementation and Performance Analysis of Interpolation Search

Array: [2, 5, 10, 15, 23, 35, 48, 60, 75, 90, 105, 120]
Searching for: 35
Found at index: 5, Comparisons: 3

Size	IS Time(ms)	BS Time(ms)	IS Comparisons	BS Comparisons
1000	0.0012	0.0009	4	8
5000	0.0034	0.0041	4	12
10000	0.0038	0.0042	5	13
50000	0.0020	0.0018	7	15
100000	0.0031	0.0051	4	16

2. Comparative Analysis of Naive, Rabin-Karp, and KMP Algorithms for String Matching

Text: AABAACAADAABAABA

Pattern: AABA

Naive -> Matches at: [0, 9, 12], Comparisons: 30

KMP -> Matches at: [0, 9, 12], Comparisons: 18

RK -> Matches at: [0, 9, 12], Comparisons: 12

Pattern	Naive	KMP	RK
AB	12492	10000	1302
ABCD	13295	10000	234
ABCDAB	13344	10012	140
ABCDABCD	13341	10012	127

3. Implementation of Kruskal's and Prim's Algorithms for Minimum Spanning Tree

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=== Kruskal's MST ===
Edge (0 - 3) Weight: 5
Edge (2 - 4) Weight: 5
Edge (3 - 5) Weight: 6
Edge (0 - 1) Weight: 7
Edge (1 - 4) Weight: 7
Edge (4 - 6) Weight: 9
Total MST Cost: 39

=== Prim's MST ===
Edge (0 - 3) Weight: 5
Edge (3 - 5) Weight: 6
Edge (0 - 1) Weight: 7
Edge (1 - 4) Weight: 7
Edge (4 - 2) Weight: 5
Edge (4 - 6) Weight: 9
Total MST Cost: 39
```

4. Implementation of Single Source Shortest Path Algorithm (Dijkstra's)

Shortest paths from vertex 0:		
Vertex	Distance	Path
0	0	0
1	3	0 -> 2 -> 1
2	1	0 -> 2
3	4	0 -> 2 -> 1 -> 3
4	7	0 -> 2 -> 1 -> 3 -> 4
5	9	0 -> 2 -> 1 -> 3 -> 4 -> 5

5. To Find Min-Max Value by Applying Divide and Conquer Technique

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Array: [3, 1, 7, 4, 9, 2, 8, 5, 6, 0]			
Min: 0, Max: 9			
D&C Comparisons: 14			
Naive Comparisons: 18			
Size	DC Comps	Naive Comps	Formula $3n/2 - 2$
10	14	18	13
100	162	198	148
1000	1510	1998	1498
10000	15902	19998	14998